

PANIMALAR GOBICHETTIPALAYAM ANNADURAI

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SUMMARY

Data scientist with **5+ years** of industry experience and **two master's degrees** in computer science and data science, building production machine learning, data and GenAI systems. Work spans recommendation and ranking systems, causal inference, counterfactual modeling, statistical modeling and hypothesis testing, LLM and RAG applications, and cloud ML platforms on Amazon Web Services (AWS) and Google Cloud Platform (GCP). Applies structured problem solving and advanced analytics on large datasets, and communicates analysis insights and actionable recommendations to executive and non-technical audiences. Currently building production AI tools at Case Western Reserve University, with prior experience at Yubi, Samsung and SymphonyAI.

TECHNICAL SKILLS

Machine Learning and AI: Machine learning, deep learning, natural language processing (NLP), recommendation and ranking systems, causal inference, counterfactual modeling, statistical modeling, econometrics and time-series modeling, forecasting, hypothesis testing, A/B testing and experimental design, sensitivity analysis, decision analysis, data mining, model evaluation, graph neural networks, ASR

GenAI and Agentic AI: LLMs, RAG, vector databases, embeddings, prompt engineering, LLM evaluation, LoRA fine-tuning, AWS Bedrock, Anthropic Claude, LangChain, LangGraph, Claude Agent SDK, agentic and multi-agent workflows, human-in-the-loop

Data and Engineering: Python, SQL, Java, Scala, PySpark, Pandas, Airflow, ETL/ELT, ML pipelines, ad-hoc data models, data modeling, large datasets, data quality and validation, REST APIs, Docker, Kubernetes, CI/CD, Git

Frameworks: PyTorch, TensorFlow, scikit-learn, FAISS

Cloud and Platforms: Amazon Web Services / AWS (S3, EC2, SageMaker, Redshift, Bedrock), Google Cloud Platform / GCP (Vertex AI, BigQuery), Azure, Snowflake

Business and Communication: Microsoft Excel, Microsoft PowerPoint, structured problem solving, advanced analytics, executive communication (written and verbal), cross-functional stakeholder management, presentations and reports for management and business customers

PROFESSIONAL EXPERIENCE

AI Solutions Developer, Case Western Reserve University, Cleveland, OH *Jan – Aug 2026*

- Built and deployed AI tools used by clinical investigators, curating the AI application catalog on the CWRU AI Playground.
- Designed the data model and ingestion process for a searchable clinical research database on BigQuery (Google Cloud Platform), making unstructured study metadata queryable for downstream analysis and ad-hoc data models over large datasets.

Data Scientist II, Yubi, Chennai, India *Jun 2020 – Dec 2024*

- Designed and deployed recommendation and ranking systems serving millions of users, improving click-through rate by 12 to 18% and driving measurable revenue growth, with impact validated through A/B testing and hypothesis testing.
- Built batch data and ML pipelines in Python and Airflow across S3, EC2, SageMaker and Redshift, reducing model refresh latency by 35%.
- Modeled raw interaction and transaction data into Redshift reporting tables with PySpark, and added automated validation and reconciliation checks to the deployment path so malformed or incomplete loads failed the pipeline rather than the dashboard.
- Owned YUCI-ASR, the in-house speech recognition model, from research direction through production release, and presented modeling tradeoffs and actionable recommendations to executive stakeholders.

Data Engineer, Samsung, Chennai, India *Nov 2019 – Jun 2020*

- Built NLP pipelines for the Samsung RCS AI Chatbot, processing high volumes of user interaction logs to improve intent classification accuracy and response relevance.
- Implemented a session-based recommender on Gated Graph Neural Networks over user interaction data, reaching 94.6% accuracy.

Deep Learning Intern, SymphonyAI, Bangalore, India *May – Nov 2018*

- Built and deployed medical named entity recognition systems on Bi-LSTM and CRF architectures at 96.8% accuracy, covering feature engineering, model training, validation and evaluation.
- Co-authored *Deep Learning with Applications Using Python* (Apress, 2018), an Amazon best-selling textbook, written

alongside the team's applied work.

RESEARCH AND MASTER'S THESIS

Feedback Loop Intervention for Long-Term Fairness in ML-Enabled Systems

May 2025 – Aug 2026

Case Western Reserve University · Advisor: Prof. Sumon Biswas

- Modeled temporal feedback loops in loan-lending, opioid risk and police allocation systems using causal graphs, Bayesian networks and constraint-based formal models, quantifying instability and safety-utility tradeoffs under policy interventions and counterfactual scenarios.
- Applied causal discovery (structural causal models, VARLiNGAM) and benchmarked LLMs (Claude, Llama 3.1) against statistical methods on accuracy and reliability for policy selection and decision analysis.
- Built automated testing pipelines to stress-test decision systems under varying intervention strategies, running sensitivity analysis to define safety guardrails and human intervention points for autonomous decisions.
- Paper submitted to **ICSE 2027**; code at github.com/pxg399/CALM.

PROJECTS

RAG-Based Intelligent Document QA System

Nov – Dec 2025

- Built an end-to-end RAG application on LangChain, FAISS and LLaMA-2 for natural language querying over domain-specific corpora, reducing the hallucination rate by 40% against a naive LLM baseline.
- Deployed as a REST API on AWS EC2 at sub-2s latency using semantic chunking, vector indexing and context-aware prompt engineering.

Fine-Tuning Diffusion Models for E-Commerce Product Image Generation

Apr – May 2025

- Fine-tuned Stable Diffusion v1.4 and SDXL with LoRA on the KREAM BLIP Captions dataset, using CLIP-based discriminator feedback and negative prompting to reach 91% brand accuracy.
- Evaluated with FID, CLIP, LPIPS and human studies, with SDXL and LoRA outperforming baseline SD models.

EDUCATION

Case Western Reserve University, Cleveland, OH

Aug 2024 – Aug 2026

MS, Computer Science (thesis track), GPA 3.67/4.00

PSG College of Technology, India

Jun 2015 – May 2020

MS (5-year Integrated), Data Science, CGPA 8.77/10.0

ACHIEVEMENTS & PUBLICATIONS

Author, *Feedback Loop Intervention for Long-Term Fairness in ML-Enabled Systems* | ICSE 2027

Jun 2026

Co-author, *Deep Learning with Applications Using Python* | SymphonyAI

Nov 2018

Best Newbie Award | Yubi | Chennai, India

Sep 2020

Yubi EPIC - New Business Ninjas | Yubi | Chennai, India

2023 – 2024

CERTIFICATIONS

- **AWS Certified AI Practitioner** (Aug 2023)
- **Computational Finance and Risk Management**, IIM Bangalore (Nov 2022)
- **Building Data Lakes on AWS** (Oct 2022)
- **The Machine Learning Pipeline on AWS** (Sep 2022)
- **Practical Data Science with Amazon SageMaker** (Aug 2022)